Bordetella parapertussis Infectious Disease — Comprehensive Disease Characterization

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Summary

Bordetella parapertussis is a Gram-negative, aerobic coccobacillus of the *B. bronchiseptica* cluster that causes **parapertussis** — a whooping-cough (pertussis-like) illness in humans and, in a genetically distinct ovine-adapted lineage, a chronic non-progressive pneumonia in sheep. In humans it colonizes the ciliated respiratory epithelium of the upper and lower airways and deploys a set of BvgAS-regulated virulence factors that it shares with *B. pertussis* (filamentous hemagglutinin, fimbriae, pertactin, the bifunctional adenylate cyclase/hemolysin toxin CyaA, dermonecrotic toxin, and tracheal cytotoxin), while **notably lacking pertussis toxin** and instead expressing an **O-antigen lipopolysaccharide** that *B. pertussis* does not make. The disease it produces is on average milder and shorter than classic pertussis and lacks the marked lymphocytosis that pertussis toxin drives — yet it is **not benign**: it produces measurable morbidity, hospitalizations, ICU admissions, and rare deaths, and in older children has been associated with *higher* odds of severe outcomes than *B. pertussis*.

The single most consequential public-health feature of this organism is that **current pertussis vaccines provide little to no cross-protection against it**. Acellular pertussis (aP) vaccines are built primarily around pertussis toxin and a handful of *B. pertussis* surface antigens, and because *B. parapertussis* neither expresses pertussis toxin nor shares the O-antigen, immunity generated by vaccination does not neutralize it. A large meta-analysis found no significant protective effect, and animal work shows aP vaccination can paradoxically *facilitate B. parapertussis* infection by dampening the innate/neutrophil response that would otherwise limit it. This has evolutionary and epidemiological implications: as pertussis-toxin-directed immunity suppresses *B. pertussis*, an ecological niche may open for the vaccine-escaping *B. parapertussis*.

This report synthesizes the mechanistic and clinical picture assembled over the investigation. Because *B. parapertussis* infection is an **acquired infectious disease with no human genetic cause**, the sections of the research template that concern causal genes, pathogenic variants, inheritance patterns, penetrance, and germline genetics are **not applicable**; the relevant "genetics" is that of the bacterial pathogen and its virulence loci, which are covered under Etiology, Mechanism, and Molecular Profiling. Ten findings were confirmed across 45 reviewed papers; all key claims below are anchored to primary literature with PMIDs and verified abstract quotes.

Key Findings

F001 — Current pertussis vaccines provide little to no protection against *B. parapertussis*

A systematic review and meta-analysis encompassing **46,533 participants** found no statistically significant protective effect of pertussis vaccination against *B. parapertussis* (PMID: 38543887). The mechanistic explanation is straightforward and well supported: acellular vaccines are constructed around *B. pertussis* antigens — chiefly **pertussis toxin (PT)**, which *B. parapertussis* does not express at all — while *B. parapertussis* produces an **O-antigen LPS** that is absent from *B. pertussis*, so there is little antigenic overlap on which cross-immunity could be built. A complementary study developing a recombinant iron-transport protein antigen explicitly concluded that *"New protective antigens are needed to improve the efficacy of current vaccines against both species"* (PMID: 28857261). Consistent with the meta-analysis, a large German efficacy trial found the multicomponent DTaP vaccine had at best a weak, non-significant point estimate of efficacy against *B. parapertussis* (31%, 95% CI −10 to 56 for cough ≥7 days) (PMID: 9417143). The take-home is that parapertussis is a genuine **vaccine-escape phenotype**, not merely an under-vaccinated population effect.

F002 — *B. parapertussis* causes clinically significant disease, with higher odds of severe outcomes than *B. pertussis* in older children

A multicentre retrospective, propensity-matched study comparing **2,554 *B. parapertussis* cases with 8,058 *B. pertussis* cases** found that, overall after matching, the odds ratios for hospitalization and critical-care admission in the *B. parapertussis* group were **1.75 (95% CI 1.46–2.10; p < 0.001)** and **1.94 (95% CI 1.28–2.93; p = 0.001)** respectively (PMID: 41314606). The

age-stratified pattern is important and nuanced: in infants under 1 year the hospitalization OR was *below* 1 (0.69, 95% CI 0.49–0.96), i.e., *B. pertussis* was more severe in the youngest, but in older age bands *B. parapertussis* was associated with markedly higher odds of hospitalization — 1–4 years OR 3.34 (2.27–4.91), 5–17 years OR 4.22 (2.62–6.79), and ≥18 years OR 1.76 (1.06–2.94). This directly refutes the notion that parapertussis is trivially mild across the board. Corroborating case-level evidence comes from a Turkish five-year hospital series in which **2 of 13 *B. parapertussis* patients died**, and the authors noted that *"infections due to B.parapertussis can also be severe and fatal"* and that most *B. parapertussis* patients were vaccinated — reinforcing both severity and vaccine escape (PMID: 40277262).

F003 — Virulence factors are BvgAS-regulated and overlap with *B. pertussis* but lack pertussis toxin

The three classical bordetellae (*B. pertussis*, *B. bronchiseptica*, *B. parapertussis*) produce a **nearly identical set of virulence factors**: adhesins (filamentous hemagglutinin FHA, fimbriae, pertactin) and toxins (the bifunctional adenylate cyclase/hemolysin CyaA, dermonecrotic toxin, tracheal cytotoxin). The abstract of the landmark pathogenesis review lists *"toxins such as a bifunctional adenylate cyclase/hemolysin, dermonecrotic toxin, tracheal cytotoxin, a B. pertussis specific pertussis toxin"* — explicitly flagging PT as *B. pertussis*-specific — and states that *"Expression of nearly all of these virulence factors is positively regulated by the products of the bvgAS locus"* (PMID: 11689354). The **BvgAS two-component signal-transduction system** is therefore the master regulator that switches the bacterium between its virulent (Bvg+) and avirulent/environmental (Bvg−) phases. The absence of PT is the central molecular distinction that explains both the milder systemic phenotype (no PT-driven lymphocytosis or islet-activation effects) and the failure of PT-based vaccines.

F004 — O-antigen LPS is a key immune-evasion/colonization factor unique to *B. parapertussis* (absent in *B. pertussis*)

The **wbm locus** genes assemble the O-antigen in *B. parapertussis* and *B. bronchiseptica* but are **absent in *B. pertussis***. In vitro, *"Complement was not activated or depleted by wild-type bordetellae expressing O antigen, but both Deltawbm mutants activated complement and were highly sensitive to complement-mediated killing"* (PMID: 12496152). Critically, the O-antigen is also required for colonization: *"The B. parapertussis Deltawbm mutant was severely defective in colonization of the tracheas and lungs of mice"* (same study). The O-antigen thus performs a **dual immune-evasion + colonization** role and is simultaneously the antigen most responsible for the vaccine gap (since *B. pertussis*-based vaccines contain no O-antigen). A second shared

adhesin, **pertactin (PRN)** — an autotransporter that is itself a component of most acellular vaccines — is required to resist neutrophil-mediated clearance: *"PRN plays a critical role in allowing Bordetella to resist neutrophil-mediated clearance"* (PMID: 20421378).

F005 — Two host-associated forms: human (whooping cough) and ovine (chronic non-progressive pneumonia in sheep)

B. parapertussis exists as two host-restricted lineages. The **ovine** lineage causes **chronic non-progressive pneumonia** in sheep and lambs and is a veterinary pathogen of economic importance. Intranasal inoculation of mice with ovine isolates *"produced a subacute bronchopneumonia morphologically similar to early lesions of naturally occurring ovine chronic non-progressive pneumonia"* (PMID: 2918110), with neutrophil/macrophage infiltration, peribronchiolar lymphoid hyperplasia, and interalveolar septal fibrosis. Electron microscopy identified the cellular targets of tissue damage: *"Electron microscopy showed damage to ciliated cells, type 1 pneumocytes and alveolar macrophages"* (PMID: 2300709). Notably, **cell-free culture fluid alone reproduced milder lesions**, implicating secreted toxins in the pathology. Co-infection with *Pasteurella haemolytica* produced synergistically more severe bronchopneumonia (PMID: 2061424). Clearance studies showed the organism is cleared faster from trachea (~100% by 48 h) than lung (~55%) (PMID: 16031626). The human and ovine lineages are genetically and phenotypically distinct despite the shared species name.

F006 — Acellular pertussis vaccination and *B. pertussis* co-infection can *facilitate B. parapertussis* infection

Two independent animal studies show that immunity or co-infection directed at *B. pertussis* can *help B. parapertussis*. In a rodent model, *"aP vaccination helped clear B. pertussis but resulted in an approximately 40-fold increase in B. parapertussis lung colony-forming units (CFUs)"*, with reduced lung inflammatory and neutrophil responses (PMID: 20200027). In a mouse co-infection model, *"Infection of the mouse respiratory tract by B. parapertussis was not only enhanced by the presence of B. pertussis, but B. parapertussis significantly outcompeted B. pertussis"* — an enhancement that required pertussis toxin and was reversed by airway macrophage depletion (PMID: 21707780). Together these suggest a mechanism whereby PT-mediated suppression of innate immunity, whether from co-infecting *B. pertussis* or (paradoxically) the immune landscape after aP vaccination, creates a permissive environment for the vaccine-escaping species.

F007 — Parapertussis is on average milder and shorter than pertussis, with measurable incidence

In an Italian acellular-pertussis efficacy cohort of **15,601 children**, 76 *B. parapertussis* infections were diagnosed, giving an incidence of **2.1 laboratory-confirmed parapertussis cases per 1,000 person-years** (PMID: 9542925). The same study found that *"Children affected by B. parapertussis infections showed a less severe clinical picture both in the duration of symptoms and in the percentage of patients affected, even when compared with vaccinated children with pertussis"* — and that *B. parapertussis* strains showed high genetic stability across time and geography. Surveillance studies consistently show *B. parapertussis* as a **minority of Bordetella detections**: 0.1% (10/7732) in China (PMID: 39002934), ~6% in Tunisia (PMID: 22313629), and ~5.9% in a 9-year Tunisian series (PMID: 30526740). The "milder on average" finding (F007) and the "higher odds of severe outcome in older children" finding (F002) are not contradictory: the *population average* is milder, but the *conditional* risk given infection in older age groups can be higher than for *B. pertussis*.

F008 — Clinical phenotype spectrum (applicable to *B. parapertussis*) with HPO terms and frequencies

Because parapertussis is clinically a whooping-cough syndrome, the phenotype spectrum mirrors pertussis, generally attenuated. In a series of 247 children with whooping cough, **paroxysmal cough occurred in 96.4% (238/247)** and **inspiratory whoop in 24.7% (61/247)**, with cyanosis, apnea and suffocation concentrated in infants under 3 months (PMID: 26757969). A Taiwan series reported whooping cough 69.6%, post-tussive vomiting 22.2%, cyanosis 59.1%, leukocytosis (WBC ≥15,000) 47.2%, lymphocytosis 44.4%, and complications in 30.4% — of which *"pneumonia was the most common (92.3%)"* (PMID: 15497010). Severe complications documented for the syndrome include severe pneumonia, encephalopathy, apnea, pulmonary hypertension, extreme leukocytosis, and seizures (PMID: 41749225). **Important nuance:** the marked leukocytosis/lymphocytosis is pertussis-toxin driven and therefore expected to be *less pronounced or absent* in *B. parapertussis*, since it lacks PT.

Phenotype → HPO mapping:

Phenotype	HPO term	Frequency (whooping-cough series)
Paroxysmal cough	HP:0002102 (Cough)	96.4%
Inspiratory whoop	HP:0031251 (Paroxysmal cough)/whoop	24.7%
Post-tussive vomiting	HP:0002013 (Vomiting)	22.2%
Cyanosis	HP:0000961 (Cyanosis)	59.1% (series-dependent)
Apnea	HP:0002104 (Apnea)	infants <3 mo
Pneumonia	HP:0002090 (Pneumonia)	most common complication (92.3% of complications)
Leukocytosis	HP:0001974 (Leukocytosis)	47.2% (PT-driven; less in parapertussis)
Lymphocytosis	HP:0100827 (Lymphocytosis)	44.4% (PT-driven; less in parapertussis)
Seizure / encephalopathy	HP:0001250 / HP:0001298	severe cases

F009 — Pathophysiology causal chain: adhesion → adenylate cyclase immune paralysis → TCT/endotoxin-mediated ciliated-cell destruction

The shared virulence factors act in sequence. The **adenylate cyclase toxin (CyaA)** targets CD11b/CD18-expressing innate immune cells — *"targeting innate immune cells which express CD11b/CD18, the cellular receptor of CyaA"* — translocates its catalytic domain into the cytosol where calmodulin activates it to produce supraphysiologic cAMP, causing cell rounding, cytoskeletal remodeling, and inhibition of migration and epithelial repair at concentrations as low as ~0.5 nM (PMID: 32392246). **Tracheal cytotoxin (TCT)**, a peptidoglycan-derived glycopeptide, synergizes with **endotoxin (LPS)** to drive iNOS/nitric-oxide production in non-ciliated airway epithelial cells, causing the selective destruction of neighboring **ciliated cells**: *"TCT and endotoxin are both important virulence factors of B. pertussis, combining synergistically to cause the specific epithelial pathology of pertussis"* (PMID: 11207540). The ovine data close the loop by showing that *B. parapertussis* **cell-free culture fluid alone** reproduces bronchopneumonia with ciliated-cell and type-1-pneumocyte damage (PMID: 2300709), demonstrating that secreted toxins — not just live bacterial invasion — mediate the tissue injury.

F010 — Diagnosis relies on IS1001 PCR (not anti-PT serology); treatment is macrolides (azithromycin first-line)

Species-specific real-time PCR on nasopharyngeal specimens identifies *B. parapertussis* by the insertion sequence **IS1001**, distinguishing it from *B. pertussis* (IS481+/ptxP+) and *B. holmesii* (hIS1001+): the triplex assay targets *"the insertion sequence IS481 and the pertussis toxin promoter region (ptxP) for B. pertussis, and the insertion sequence IS1001 for B. parapertussis"* and detects ~1 CFU per reaction (PMID: 20718721; corroborated by PMID: 21393459, PMID: 22313629, PMID: 24586447). **Anti-pertussis-toxin serology is *B. pertussis-specific and useless for B. parapertussis*** because the organism lacks PT. For treatment, susceptibility testing showed *"The most active antibiotic against all strains was azithromycin (MIC90 = 0.06 mg/l)"*, with erythromycin also effective (MIC90 ~0.25 mg/L) and no significant resistance drift between 1968 and 1997–99 isolates (PMID: 11757403). Macrolides — **azithromycin first-line** — are therefore the mainstay, mirroring pertussis therapy.

Section-by-Section Disease Characterization

1. Disease Information

- **Overview:** Parapertussis is an acute respiratory infection caused by *B. parapertussis*, presenting as a whooping-cough (pertussis-like) illness in humans, generally milder and shorter than classic pertussis. A separate ovine-adapted lineage causes chronic non-progressive pneumonia in sheep.
- **Identifiers:** MONDO:0001353; MeSH "Whooping Cough" / "Bordetella parapertussis"; ICD-10 **A37.1** (Whooping cough due to *Bordetella parapertussis*); ICD-11 **1C12** (Whooping cough) with pathogen specification. NCBI Taxonomy **txid519**. No OMIM/Orphanet entry (not a Mendelian disease).
- **Synonyms:** Parapertussis; *B. parapertussis* whooping cough; parapertussis pertussis-like syndrome.
- **Data source:** Aggregated disease-level resources plus clinical case series and surveillance cohorts (EHR-derived retrospective studies such as

  P 41314606

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  Not a genetic/EHR-variant resource.

2. Etiology

- **Causal factor:** Infectious — the Gram-negative coccobacillus *B. parapertussis* (BvgAS-regulated virulence program; F003). No human genetic cause.
- **Risk factors (environmental/host):** Young age (infants at highest risk of severe outcomes; P 41314606), close respiratory contact (household transmission; siblings and mothers as sources, P 22313629, P 39565788), lack of/incomplete vaccination for *B. pertussis* offers **no protection** here (F001), crowding, and co-circulation of *B. pertussis*. **Human genetic risk/protective factors: not applicable / none established.**
- **Gene–environment interactions:** The key interaction is at the pathogen–host-immunity level: prior *B. pertussis*-directed immunity or co-infection can facilitate *B. parapertussis* (F006). No human GxE evidence.
- **Protective factors:** Whole-cell pertussis vaccines and natural immunity provide partial/uncertain cross-protection at best; O-antigen-based or novel-antigen vaccines (e.g., recombinant iron-transport protein, P 28857261; live-attenuated BPZE1, P 35196124) are experimental.

3. Phenotypes

See **F008** table. Phenotype type: symptoms/signs (paroxysmal cough, whoop, post-tussive vomiting, cyanosis, apnea) and laboratory abnormalities (leukocytosis/lymphocytosis, generally *less* than pertussis due to absent PT). **Onset:** childhood-predominant, all ages affected. **Severity:** mild-to-moderate on average, severe in a minority (F002, F007). **Progression:** self-limited, episodic paroxysms over days–weeks. **QoL impact:** cough paroxysms, sleep disruption, post-tussive vomiting; hospitalization in severe cases.

4. Genetic/Molecular Information

Not applicable in the human-genetic sense — parapertussis has no causal human genes, pathogenic variants, inheritance pattern, penetrance, or chromosomal abnormalities. The relevant molecular information is **bacterial**: - **Virulence loci:** *bvgAS* (master regulator), *cyaA* (adenylate cyclase toxin), *fhaB* (FHA), *prn* (pertactin), *fim* (fimbriae), *dnt* (dermonecrotic toxin), *wbm* locus (O-antigen biosynthesis, unique vs *B. pertussis*), TCT (peptidoglycan-derived, released during cell-wall turnover). **Pertussis toxin operon (ptx) is absent/non-functional.** - **Comparative genomics:** *B. parapertussis* retains functional genes and expresses proteins absent or pseudogenized in *B. pertussis* (proteomic evidence, including O-antigen and stress-resistance/virulence determinants;).

P 31325608

Human *B. parapertussis* strains show high genetic stability ( ).

P 9542925

The BatB immunoglobulin-binding autotransporter is deleted/truncated in human *B. parapertussis* relative to *B. bronchiseptica* ( ).

P 18426869

5. Environmental Information

- **Infectious agent:** *B. parapertussis* (NCBI txid519), *B. bronchiseptica* cluster.
- **Transmission:** Respiratory droplets, human-to-human (human lineage); host-restricted ovine lineage in sheep.
- **Environmental/lifestyle factors:** Crowding, close contact; no established toxin/pollution/dietary etiology.

6. Mechanism / Pathophysiology — Ordered Causal Chain

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1. Inhalation of B. parapertussis → deposition on ciliated respiratory epithelium (trachea, bron
        |
2. BvgAS two-component system senses host environment → switches to Bvg+ (virulent) phase
   → transcribes virulence factors (INFERRED to mirror B. pertussis regulation; <a href="https:/
        |
3. Adhesins (FHA, fimbriae, pertactin) + O-antigen LPS → firm attachment & colonization of cilia
   → pertactin resists neutrophil clearance (<a href="https://pubmed.ncbi.nlm.nih.gov/20421378/"
        |        └── BRANCH (immune evasion): O-antigen blocks complement activation/deposition
        |
4. Adenylate cyclase toxin (CyaA) binds CD11b/CD18 on innate immune cells → delivers catalytic c
   → calmodulin-activated supraphysiologic cAMP → phagocyte paralysis, impaired migration & epit
        |
5. Tracheal cytotoxin (TCT) + endotoxin (LPS) synergize → induce iNOS/nitric oxide in non-ciliat
   → selective destruction of neighboring ciliated cells (<a href="https://pubmed.ncbi.nlm.nih.g
        |
6. Loss of ciliated cells + mucociliary clearance failure + secreted-toxin injury (reproduced by
   → airway inflammation, epithelial damage, mucus accumulation
        |
7. Clinical manifestation → paroxysmal cough, whoop, post-tussive vomiting; in severe cases pneu
   apnea (infants), and rare death (<a href="https://pubmed.ncbi.nlm.nih.gov/26757969/" rel="noo
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- **Upstream drivers:** BvgAS regulation, adhesion, immune evasion (O-antigen, pertactin, CyaA).
- **Downstream effectors:** TCT/LPS-mediated ciliated-cell destruction, inflammation, cough reflex.
- **Key distinction from pertussis:** absence of PT means no PT-driven systemic effects (lymphocytosis, hypoglycemia, histamine sensitization) → milder average phenotype.
- **GO terms:** GO:0009405 (pathogenesis), GO:0044179 (hemolysis in another organism), GO:0052572 (response to host immune response), GO:0006954 (inflammatory response), GO:0071731 (response to nitric oxide). **CL terms:** CL:0000064 (ciliated cell), CL:0002370 (respiratory ciliated epithelial cell), CL:0002062 (type I pneumocyte), CL:0000583 (alveolar macrophage), CL:0000775 (neutrophil).

7. Anatomical Structures Affected

- **Primary organs:** Trachea (UBERON:0003126), bronchus (UBERON:0002185), lung (UBERON:0002048), nasopharynx (UBERON:0001728). Respiratory system (UBERON:0001004).

- **Secondary:** Lung parenchyma in pneumonia; rare systemic/CNS effects (encephalopathy — brain, UBERON:0000955).
- **Tissue/cell level:** Respiratory epithelium (epithelial tissue); target cells = ciliated epithelial cells (CL:0000064), type I pneumocytes (CL:0002062), alveolar macrophages (CL:0000583), with neutrophil infiltration.
- **Subcellular:** Cytoplasm (cAMP dysregulation; GO:0005737), plasma membrane receptor CD11b/CD18 (GO:0005886), bacterial outer membrane (O-antigen; GO:0009279).
- **Lateralization:** Bilateral airway involvement.

8. Temporal Development

- **Onset:** Any age; childhood-predominant. Incubation ~7–10 days. Acute onset with catarrhal stage → paroxysmal stage → convalescence, generally shorter than pertussis (F007).
- **Progression:** Self-limited, episodic paroxysms; most resolve over weeks. Severe/complicated course in a minority (infants, or older children per F002).
- **Duration:** Self-limited (not chronic in humans). Ovine lineage causes *chronic* non-progressive pneumonia in sheep (F005).
- **Remission:** Spontaneous resolution typical; antibiotics shorten infectivity and may attenuate course if given early.

9. Inheritance and Population

- **Inheritance: Not applicable** — infectious, not heritable. Penetrance/expressivity/anticipation/founder-effect/carrier-frequency concepts do not apply.
- **Epidemiology:** Incidence ~**2.1 laboratory-confirmed cases per 1,000 person-years** in one pediatric cohort (F007, P 9542925). Represents a **minority of Bordetella detections** — 0.1% (China, P 39002934) to ~6% (Tunisia, P 22313629 / 30526740). Global distribution wherever pertussis circulates.

- **Demographics:** Affects children of all ages; infants at highest risk of severe disease; older children/adults also affected. No strong ethnic predisposition (host-genetic factors not established). Sex ratio not clearly skewed.

10. Diagnostics

- **Gold standard: Species-specific real-time PCR targeting IS1001** on nasopharyngeal specimens (F010,

  P 20718721

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  B. parapertussis = IS481−/IS1001+/ptxP−.
- **Culture:** Possible on Bordet-Gengou / Regan-Lowe media but low sensitivity.
- **Serology: Anti-PT serology is NOT useful** (organism lacks PT).
- **Laboratory:** Leukocytosis/lymphocytosis are typically **less pronounced** than in pertussis (PT-dependent).
- **Imaging:** Chest radiograph for suspected pneumonia complication.
- **Differential diagnosis:** *B. pertussis*, *B. holmesii* (hIS1001+), *B. bronchiseptica*, RSV and other viral respiratory infections (RSV frequently co-tested,

  P 21393459

  ),
  Mycoplasma, chlamydial pneumonitis.

11. Outcome / Prognosis

- **Prognosis:** Generally good — most cases self-limited and milder than pertussis (F007).
- **Mortality:** Low but non-zero; deaths reported (e.g., 2/13 in one series,

  P 40277262

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- **Severe outcomes:** Higher odds of hospitalization/ICU than *B. pertussis* in older children (F002); infants at risk of apnea/pneumonia.
- **Complications:** Pneumonia (most common), apnea, rarely encephalopathy/seizures, pulmonary hypertension in severe whooping-cough-syndrome cases
  (
  P 15497010

  P 41749225

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- **Prognostic factors:** Young age, comorbidities, extreme leukocytosis, pulmonary hypertension (adverse markers in the broader whooping-cough syndrome).

12. Treatment

- **First-line pharmacotherapy: Macrolides — azithromycin** (most active, MIC90 ~0.06 mg/L), erythromycin, clarithromycin (F010, P 11757403).
  Trimethoprim-sulfamethoxazole as alternative. NCIT: Azithromycin (C1029), Erythromycin (C376), Clarithromycin (C614).
- **Note on macrolide resistance:** High-level macrolide resistance is well documented in *B. pertussis* in some regions (P 33021592);
  B. parapertussis has not shown comparable resistance drift, but β-lactams (e.g., piperacillin) are potential alternatives.
- **Supportive care:** Hydration, respiratory support, oxygen; ICU/mechanical ventilation for severe cases.
- **Advanced/experimental:** Live-attenuated BPZE1 and Bbvac candidates confer cross-protection in models (P 35196124; P 22814407);
  recombinant iron-transport-protein antigen protective in mice (P 28857261).
  No gene/cell/RNA therapy applicable.

13. Prevention

- **Primary prevention: Current pertussis vaccines do NOT reliably prevent parapertussis** (F001). This is the central prevention gap. Whole-cell vaccines may offer marginal cross-protection; acellular vaccines essentially none and may facilitate infection (F006).
- **Novel vaccine strategies:** O-antigen-inclusive or shared-antigen vaccines, live-attenuated BPZE1/Bbvac (Th17/Th1-mediated cross-protection), recombinant iron-transport protein.

- **Secondary/tertiary:** Early PCR diagnosis, macrolide treatment to reduce transmission, isolation of cases, household contact tracing.
- **Public health:** Maternal vaccination protects infants against *B. pertussis* (P 29346585) but **not** against *B. parapertussis*; surveillance with species-specific PCR is essential to detect vaccine escape.

14. Other Species / Natural Disease

- **Taxonomy:** *B. parapertussis* NCBI **txid519**; *B. bronchiseptica* cluster.
- **Ovine natural disease:** Chronic non-progressive pneumonia in **sheep** (*Ovis aries*, NCBI txid9940) — a distinct host-adapted lineage of veterinary importance (F005). Recapitulated in mice; synergy with *Pasteurella haemolytica* (P 2918110 P 2300709 P 2061424 P 16031626).
- **Comparative pathology:** Both human and ovine lineages target ciliated epithelium via shared toxins; ovine disease is chronic/pneumonic rather than the acute cough syndrome of humans.
- **Zoonotic potential:** Human and ovine lineages are host-restricted; no significant documented zoonotic cross-over.

15. Model Organisms

- **Mouse** (*Mus musculus*): Primary model — intranasal challenge reproduces colonization, bronchopneumonia (ovine isolates), and the vaccine-facilitation and co-infection phenomena (P 2918110 P 2300709 P 20200027

P 21707780). P 12496152 Recapitulates ciliated-cell/type-1-pneumocyte damage and colonization defects of Δwbm mutants.

- **Rat:** Rodent bordetellosis model used for aP-vaccination facilitation study (P 20200027).
- **Baboon:** Used for pertussis maternal-vaccination studies (P 29346585) — relevant to comparative vaccine biology.
- **In vitro:** Epithelial cell invasion assays for CyaA (P 32392246); complement-killing assays for O-antigen mutants (P 12496152).
- **Recapitulation/limitations:** Mouse models reproduce colonization, tissue tropism, and immune evasion well, but the human paroxysmal-cough/whoop phenotype is not fully modeled; ovine natural disease differs from the human syndrome.

Mechanistic Model / Interpretation

The unifying narrative is that ***B. parapertussis* is a "pertussis-toxin-null, O-antigen-positive" sibling of *B. pertussis***, and these two molecular differences explain nearly its entire distinctive clinical and public-health profile:

Feature	*B. pertussis*	*B. parapertussis*	Consequence
Pertussis toxin (PT)	**Present**	**Absent**	Milder systemic disease, less lymphocytosis; anti-PT serology & PT-based vaccines useless for parapertussis
O-antigen LPS (wbm)	**Absent**	**Present**	Complement resistance + colonization; a vaccine target the current vaccines lack
Shared factors (FHA, fimbriae, pertactin, CyaA, TCT, DNT)	Present	Present	Common adhesion / immune-evasion / ciliated-cell-killing mechanism
BvgAS regulation	Yes	Yes	Coordinated virulence expression
PCR signature	IS481+ / ptxP+	IS1001+	Species-specific diagnosis

The **O-antigen is simultaneously the reason for immune evasion AND the reason for vaccine escape** — an elegant, if unfortunate, convergence. Meanwhile the **absence of PT** softens the systemic disease but does *not* make it benign, because the ciliated-epithelium destruction is driven by TCT + endotoxin + CyaA, all of which *B. parapertussis* retains. The evolutionary corollary (F006) is that PT-directed vaccination, by suppressing *B. pertussis* and its innate-immune-dampening PT, may **open an ecological niche** for the vaccine-escaping *B. parapertussis* — a hypothesis supported by animal facilitation data and consistent with the observed persistence of parapertussis in vaccinated populations.

Evidence Base

PMID	Paper (abbrev.)	Supports	Evidence type
38543887	Pertussis vaccines scarcely protect vs *B. parapertussis*	F001	Meta-analysis (human)
28857261	Recombinant iron-transport protein vaccine	F001, F004	Model organism
41314606	Clinical outcomes vs *B. pertussis*	F002	Human retrospective
40277262	5-year hospital series (deaths)	F002	Human case series
11689354	Mechanisms of Bordetella pathogenesis	F003	Review
12496152	Role of O-antigen in respiratory infection	F004	Model organism / in vitro
20421378	Pertactin resists neutrophil clearance	F004	Model organism
2918110	Ovine pneumonia in mice	F005	Model organism
2300709	Cell-free extract pneumonia	F005, F009	Model organism
20200027	aP vaccination facilitates *B. parapertussis*	F006	Model organism
21707780	*B. pertussis* enhances *B. parapertussis*	F006	Model organism
9542925	Epidemiology/clinical/molecular	F007	Human cohort
26757969	247 children whooping cough	F008	Human case series
15497010	Taiwan pertussis series	F008	Human case series
32392246	CyaA epithelial invasion	F009	In vitro
11207540	TCT + endotoxin NO signaling	F009	In vitro
20718721	Triplex PCR IS1001	F010	Diagnostic
11757403	Macrolide susceptibility	F010	In vitro

Supporting/context papers:

P 39002934

P 22313629

P 30526740

P 21393459

P 24586447

(surveillance/diagnostics); 9417143 (DTaP efficacy); 31325608, 18426869 (comparative proteomics/genomics); 35196124, 22814407 (experimental cross-protective vaccines); 29346585 (maternal vaccination); 33021592 (macrolide resistance in *B. pertussis*); 16031626, 2061424 (ovine model).

Limitations and Knowledge Gaps

1. **Not a genetic disease:** Template sections on causal human genes, pathogenic variants, inheritance, penetrance, and germline genetics are inapplicable; the "molecular genetics" here is bacterial.
2. **Much mechanistic data is extrapolated from *B. pertussis*.** CyaA and TCT mechanisms (F009) are best characterized in *B. pertussis*; while the shared-factor evidence (F003) and cell-free-fluid ovine data (F005/F009) make extrapolation reasonable, *B. parapertussis*-specific quantitative mechanistic data are sparse.
3. **Animal-model reliance:** Vaccine-facilitation and O-antigen findings (F004, F006) are from mouse/rat models; human confirmation of the facilitation phenomenon is lacking.
4. **Underdiagnosis:** Because many labs use IS481-only PCR or anti-PT serology, *B. parapertussis* is likely under-detected, biasing incidence estimates downward.
5. **Severity heterogeneity:** The apparent contradiction between "milder on average" (F007) and "higher odds of severe outcome in older children" (F002) needs prospective, age-stratified study to resolve confounders (testing bias, comorbidity).
6. **No human genetic susceptibility/protective factors** have been established.

Proposed Follow-up Experiments / Actions

1. **Prospective, species-specific surveillance** using triplex IS481/ptxP/IS1001 PCR to obtain unbiased incidence and severity estimates across age strata, resolving the F002/F007 tension.
2. **Human immuno-epidemiology of vaccine facilitation:** test whether aP-vaccinated individuals show altered susceptibility to *B. parapertussis* (F006), currently shown only in rodents.
3. **O-antigen-based or shared-antigen vaccine development** (building on P 28857261 P 35196124) with head-to-head cross-protection testing against both species.
4. ***B. parapertussis*-specific mechanistic studies** of CyaA and TCT-mediated ciliated-cell killing to confirm extrapolations from *B. pertussis* (F009).
5. **Genomic monitoring** of human *B. parapertussis* lineages for antigenic/antibiotic-resistance drift (baseline: high stability, P 9542925; no macrolide drift, P 11757403).
6. **Clinical guidance update** emphasizing that anti-PT serology cannot diagnose parapertussis and that macrolide (azithromycin) therapy plus IS1001 PCR is the correct standard.

Report compiled from 10 confirmed findings across 45 reviewed papers. Evidence types are labeled human clinical, model organism, in vitro, or review throughout. This disease is an acquired bacterial infection; human-genetic, inheritance, and germline sections are not applicable and are marked as such.
